

Low-Cost Initial Test (India, core ≈ USD 10,000)

Does a candidate plasticity-promoting treatment (FGF17 + 7,8-DHF) combined with a structured learning environment raise performance across different cognitive tasks in rats? A staged feasibility study, run by an Indian host laboratory under remote oversight, that informs whether a larger experiment is justified

Researcher / sponsor	Solano Moreira Furlan (remote; the host laboratory runs the experiment)
Document	India initial-test protocol, version 3 (round 2 of 3 applied)
Date	9 October 2026
Status	Not approved by an ethics committee. All reviewer verdicts so far: "approve with changes". Prices are estimates, not quotes. This PDF is intended to be handed to the host laboratory; section 10 lists what the host must complete (SOPs, quotes, approvals).

Executive summary

Question. Does a candidate plasticity-promoting treatment — FGF17 (the factor identified in young cerebrospinal fluid; Iram 2022 [1]) plus 7,8-DHF (a putative TrkB/BDNF-pathway agonist whose mechanism is disputed [2–4]) — combined with a structured enriched learning environment (ELE) raise the performance of healthy young rats across three different cognitive tasks, including one whose dimensions are not practised in the ELE (set-shifting), compared with untreated peers? Assumption to confirm: "crf" in the request was read as CSF.

What this is. A small, staged feasibility, variance and direction study: 64 rats in the main experiment (about 10–11 completers per group) and 124 rats in total. It informs — it does not confirm — whether a larger, powered experiment is justified. It cannot show "intelligence", neuroplasticity or any human benefit.

Staging. Phase 0a (engagement and reagent validation in 36 satellite rats) must pass before Phase 0b (the behavioural study) is committed. Phase 0c (EdU/Olig2 histology) is a deferred add-on: the brain hemispheres are banked at the end of 0b and stained when funds allow.

Budget (honest). Core (0a + 0b, tissue banked) ≈ USD 9.6k base, ≈ 8.1k optimistic (DSIR customs concession, low animal per-diem), ≈ 11.5k pessimistic. Phase 0c adds ≈ 1.7k (full programme ≈ 11.3k base). USD 10k therefore covers the core in the base and optimistic cases, not in the pessimistic case, and does not include any hourly equipment fees the host may charge (section 9).

Long-term goal (defined by the researcher, not tested here): cognitive gain in people with learning disabilities / altered connectivity and in older adults with Alzheimer's disease, by supporting the repair of neuronal connections. If this study goes well, the next experiment (powered factorial, Phase 1; optogenetics, Phase 2) is described in section 2, together with the known risks.

1. Aim, hypotheses and what the data can claim

- **Primary aim (feasibility):** estimate completion rate, variance and direction of cognitive performance in a 2×2 of drug package × environment; establish reagent activity and pathway engagement.
- **Exploratory hypothesis:** D+E (drug package + enrichment) performs better than control peers (C) and than either component alone.
- **Transfer:** the set-shifting task is the primary transfer readout (its stimulus dimensions — odour and digging medium — are excluded from the ELE). Hebb-Williams (problem solving) and the water maze are spatial; the ELE contains tunnels, climbing and wheels, so these two are not transfer tasks and are flagged as spatial.
- **Allowed claims:** feasibility, attrition, variance, direction with 95% interval of D+E – C overall and for set-shifting alone, pathway engagement (ERK), descriptive structure if Phase 0c is run — for the drug package as a whole.
- **Not allowed:** "smarter", "general intelligence" or a g factor [8,9]; neuroplasticity; single-agent claims; TrkB-specific claims (pERK shows pathway engagement, not TrkB); human or disease claims; replication of Iram 2022 (aged-mouse rescue; this asks about enhancement in healthy young adults, which no study found has tested).

2. Long-term goal, next experiment (including optogenetics) and translational roadmap

Recommended wording (translational reviewer, adopted):

"Our long-term aim is to test whether combining a plasticity-promoting biological treatment (an FGF17-type factor plus a validated TrkB/BDNF-pathway agent) with structured enriched learning can produce measurable cognitive benefit in humans. We pursue two **separate** programmes with distinct biology, models, ages and ethics: (a) neurodevelopmental conditions with altered connectivity (learning disabilities), and (b) age-related cognitive decline and early Alzheimer's disease. We hypothesise that supporting synaptic and myelin plasticity is **one possible contributor** to such benefit; this is a hypothesis to be demonstrated with structural and functional evidence, not an assumption, and it is not addressed by the present study. Prior neurotrophin programmes in Alzheimer's have not shown benefit (including a direct-delivery AAV-NGF trial), brain delivery of protein factors is a major obstacle, and both candidate agents face unresolved questions (human CNS delivery of FGF17; whether 7,8-DHF engages TrkB). Each stage proceeds only on pre-registered target-engagement, structural and safety gates."

Stage	Content	Gate to proceed
0 - this study	0a engagement/reagents → 0b behaviour in healthy young rats → 0c histology (deferred add-on).	Section 8.
1 - NEXT EXPERIMENT, Phase 1: powered rat confirmation	Both sexes; full factorial separating FGF17, 7,8-DHF and enrichment (and exercise); sustained infusion by pump; dose-response; durable vs on-drug effects; TrkB and FGFR necessity tests; synaptic readouts (PSD-95/synaptophysin on banked tissue, LTP).	Replicated effect on a pre-registered primary; structural effect.
1 - NEXT EXPERIMENT, Phase 2: optogenetics (only if Phase 1 is positive)	Question: how does controlled neuronal activity interact with FGF17 and with 7,8-DHF-driven plasticity? Candidate designs (to be designed and reviewed in their own round, not specified here): (a) activity × plasticity-opening — optogenetic stimulation of a defined circuit (e.g. CaMKIIa-driven channelrhodopsin in prefrontal or hippocampal pyramidal neurons) during or after learning, with and without FGF17/7,8-DHF, asking whether activity amplifies oligodendrocyte-precursor proliferation and myelination (precedent: activity-dependent myelination, Gibson 2014 [10]) and synaptic plasticity; (b) necessity — inhibitory opsins during learning or consolidation to test whether the treatment effect depends on circuit activity; (c) cell-type targeting of oligodendrocyte-lineage cells, which needs Cre lines that are mostly mouse (a species switch or AAV-based rat strategy would be required). Needs AAV vectors (biosafety approval), fibre implants, calibrated light sources, opsin-negative/fluorophore-only and light-only controls (heating and visual artefacts), and histological verification of opsin expression and fibre placement. It is beyond this budget, needs its own funding, equipment and expertise, and carries extra surgical burden.	Phase 1 positive; opsin tools validated in the host lab; own ethics approval.
2 - disease models (separate tracks)	Aged rodents (Iram-type rescue); Alzheimer's models (5xFAD, APP/PS1, TgF344-AD) — amyloid-model rescue has repeatedly failed to translate; neurodevelopmental models at developmentally appropriate ages: Fmr1 knockout (fragile X) and Down-syndrome models (Ts65Dn is being superseded) as separate tracks.	Benefit with engagement and structural evidence.

Stage	Content	Gate to proceed
3 – drug development	Clinically deliverable form (intranasal, intrathecal or engineered/receptor-selective agent — unproven); a validated TrkB-pathway drug with human-relevant PK; two-species toxicology; FGF-class risks (mitogenesis/tumour promotion, angiogenesis); regulatory plan (CDSCO New Drugs and Clinical Trials Rules 2019; ICMR ethics; US/EU IND).	Delivery solved; clean toxicology.
4 – human studies	Phase 1 safety/PK; biomarkers (MRI myelin water fraction/DTI, EEG, cognitive tests); structured learning input as digital training; Phase 2 in MCI/early AD and, separately, in neurodevelopmental populations (DCGI scrutiny; assent plus proxy consent; independent data-safety monitoring).	Safety and biomarker signal.

Hard barriers that must be stated: no validated way exists to deliver FGF17 to the human brain; the mitogenic and angiogenic risk of an FGF-class CNS agent may never clear a paediatric risk-benefit assessment for a non-fatal indication such as a learning disability, and enhancement in people without disease carries a higher ethical bar than treatment of disease; 7,8-DHF is reported not to activate TrkB reliably in cell assays [2–4]; no human trials of FGF17 or 7,8-DHF were found (verify); healthy-animal enhancement may not transfer to disease or humans. The FGF23/phosphate effect is not the relevant class risk (that is a different, endocrine FGF).

What this study must deliver to earn Stage 1: reagent activity and FGF17 pathway engagement (Phase 0a); a direction on the overall and set-shifting contrasts (0b); and, if Phase 0c is run, an EdU+/Olig2+ direction. Covariation of structure with behaviour cannot be judged at this n.

3. Design and animals

Young adult male Wistar rats (about 8–10 weeks at surgery) from a CPCSEA-registered source (ICAR-IVRI Izatnagar INR 350; NISER INR 300–500 plus transport; NIN's animal unit is being reorganised — confirm supply directly). Males only. Randomised by baseline weight and source litter by the sponsor, who also holds the allocation and drug-code key (section 10); **each wave contains 8 rats per arm**; coded syringes/vials; surgeon, injector and behaviour operators blind to drug; the environment arm cannot be fully blinded, so ELE handlers and test-day scorers are different people, all behaviour is video-recorded under coded IDs and scored blind, 20% are re-scored for inter-rater agreement (ICC), and apparatus is cleaned between rats.

Group	Content	Enrolled
Main cohort	4 arms × 16: C (vehicle + standard), E (vehicle + ELE), D (FGF17 + 7,8-DHF + standard), D+E; 2 waves × 8 per arm	64
Engagement satellite – FGF17	Vehicle / 15 / 30 / 60 min after an ICV bolus, 6 each; rapid dissection and snap-freezing of periventricular and hippocampal tissue; pERK/total ERK	24
Engagement satellite – DHF	Vehicle vs 7,8-DHF (6 each), 45 min after the 7th daily dose; pERK/total ERK (descriptive)	12
Partner rats	12 unoperated partners (6 for ELE arenas, 6 for control arenas; never swapped; aggression pre-screened; ≈5 rats per partner per condition)	12
Practice/pilot	6 cannula/dye placement and EdU labelling checks; 6 vehicle-tolerability checks (7,8-DHF vehicle × 7 days)	12
Total	Cannulated rats: 64 + 24 + 6 = 94	124

Enrolment is 16 per arm because ICV-cannula attrition in rats is commonly 30–40%; the plan assumes about 62–69% completion (10–11 per arm). Welfare: the combined burden (cannula for ~7 weeks, 53 daily injections, food restriction, swimming) is classified as moderate severity, with a scoresheet and humane endpoints (>15% below growth-adjusted weight target, injection-site lesions, cannula infection, hypothermia rule); the IAEC will be asked for a pre-submission review of severity. Pre-specified exclusions: dye/histology-confirmed ventricular placement, humane endpoints, swim speed <50% of cohort median. Housing: pair-housed with a perforated divider if the IAEC permits, otherwise single-housed, identically in all arms. **Power (honest).** With N=40–44 completers an interaction contrast reaches 80% power only for $f \approx 0.43$ – 0.46 , and a pairwise D+E vs C comparison only for $d \approx 1.3$ (unadjusted); power at $f=0.25$ is 0.34 – 0.37 . A cross-species enrichment meta-analysis reports $g \approx 0.44$ in rodents (an unpublished dissertation [15]), and healthy young rats are near ceiling on simple tasks, so a null result is likely even if a modest true effect exists. Pilot effect sizes must not be used to power the next experiment from the point estimate [12,13].

4. Interventions

Component	Specification
FGF17 (days 0–6)	Recombinant mouse FGF17 (mouse–rat sequence alignment documented before ordering). Order one 25 µg vial first, bioassay it, then order the rest. Chronic guide cannula to the right lateral ventricle (scaled coordinates; dye + histology). Four boluses on days 0, 2, 4, 6: 3 µl of ~100 µg/ml (0.3 µg each; 1.2 µg per rat) in aCSF + 0.1% BSA + heparin 10 µg/ml, awake, at 0.5–1 µl/min with the injector left 2–3 min; aliquots thawed fresh each session. Vehicle boluses in all other rats. <u>Delivery gap:</u> Iram 2022 infused for about a week with an osmotic pump (Alzet 1007D delivers ≈84 µl); four boluses total 12 µl, about 7× less volume and pulsatile. The delivered dose was not retrievable; the Methods must be read (PMC copy; ID to be confirmed) before IAEC submission, and until then FGF17 is exploratory and a null is confounded with delivery. Cannula capped after day 6.
7,8-DHF	5 mg/kg IP daily, days 0–52 (53 injections; rotated sites; fixed time after testing/ELE). Rat precedent is 7 days to 4 weeks at 5 mg/kg with 17–30% DMSO vehicles; 53 days is an extrapolation. Vehicle: 5% DMSO in corn oil or CMC-Na suspension (not ≥10% DMSO), chosen in the 6-rat tolerability check; identical in all arms.
Enriched learning environment (ELE)	Days 0–52, one 60-min pair session/day (test rat + partner) after testing, in the dark phase under red light; SOP with photos to be completed by the host before IAEC submission (arena size, object list ≥3 new objects/day, latch/string-pull/lever/chew puzzle devices on a six-level ladder, running wheels, tunnels/climbing, small fixed-calorie rewards deducted from the daily ration; engagement minutes and ladder progression logged). <u>Excluded from the ELE: odour discrimination and digging puzzles.</u> Control arena: same partner, handling and duration, empty. Wheel running and spatial experience are declared confounds.
EdU (banked label)	Start at 50 mg/kg IP on days 2, 4, 6 (25 mg/kg is below any precedent found; labelling tested in the practice group). Given in the main cohort so that the label is banked; staining is Phase 0c.

5. Timeline and workload

Days	Activity
–14 to –8	Arrival and 7-day acclimatisation/quarantine.
–7 to –1	Cannula surgery with analgesia; recovery; handling and arena habituation.

Days	Activity
0–6	FGF17/vehicle boluses (0, 2, 4, 6); EdU (2, 4, 6); ELE/control sessions and 7,8-DHF/vehicle begin on day 0.
14–21	Gradual food restriction to 85–90% of a growth-curve-adjusted target (humane rule: >15% below target); held through day 45.
22–36	Task 1: attentional set-shifting (simple/compound discrimination, intradimensional and extradimensional shifts). Training to criterion with a 7-day cap, staggered; tests ≤ 4 rats/day with two operators (2–4 h per rat). Score: log trials-to-criterion at the extradimensional shift with the intradimensional shift as covariate; rats not reaching it are assigned cap+1 (sensitivity: rank/censored analysis).
37–45	Task 2: Hebb-Williams maze — 3 days adaptation, then 6 investigator-chosen problems (a subset; not validated; the standard battery is 6 practice + 12 test problems) in a fixed order, 2–3 trials per problem averaged; error zones defined in advance; split-half reliability reported.
46	Return to ad libitum feeding.
46–52	Task 3 (last; non-appetitive): water maze, 25 ± 1 °C, core temperature before/after, warm rest cage, ≥ 5 min between trials; 5 acquisition days (46–50), probe 24 h later (51; mean proximity and quadrant time), visible-platform test (52); swim speed and floating logged.
53	Euthanasia (anaesthesia then decapitation) and hemisection: one hemisphere immersion-fixed in 4% PFA 48 h (banked, then cryoprotected and stored); the other snap-frozen (dorsal hippocampus and corpus callosum dissected on ice; banked at -80 °C).

Staffing and waves. Per wave, each day needs 32 pair-sessions (16 ELE + 16 control) in 4 arenas (≈ 8 arena-hours, two staff blocks) plus daily injections, while test days need two testing operators. A **minimum of three trained operators** is required to run the two waves 4 weeks apart (in-life ≈ 13.5 weeks). If three operators are not available the waves are run sequentially, ≥ 8 weeks apart (in-life ≈ 17.5 weeks). A per-day time-budget table is part of the SOP pack (section 10). The fixed task order (set-shifting → Hebb-Williams → water maze) is declared; ELE after testing is a post-training consolidation window, declared.

6. Endpoints and tissue

Primary analysis (per-task): the three task scores are standardised with the grand mean and the pooled within-arm SD estimated in-sample (no prior data exist), winsorised on residuals at ± 3 , signs aligned. They are analysed in one linear mixed model (task as a repeated factor, random intercept per animal; arm contrasts averaged over tasks), using all animals with ≥ 2 tasks, so informative attrition is handled under the model assumptions. The mean of the three z-scores (an **aggregate index**, not a general factor) is a descriptive companion; complete-case analysis is a sensitivity analysis. Contrasts: D+E – C overall, D+E – C for set-shifting alone (the transfer readout), and the index without the water maze. Covariate: baseline body weight only (wheel running is zero in two arms and body weight after restriction is post-randomisation, so neither is in the primary model; both are described). Completion by arm and task is reported with a pre-specified imbalance flag (>20 percentage points). The **inter-task correlation matrix is computed within arm from residuals** and is descriptive.

Block	Readouts	n
Behaviour	Three tasks; blinded video scoring.	all completers
Banked tissue (every main-cohort animal)	Immersion-fixed hemisphere (EdU/Olig2 $\pm$ later CC1; corpus callosum primary region, dentate/CA1 secondary) and snap-frozen hemisphere (PSD-95/synaptophysin and other westerns are deferred to Stage 1 or to a funded add-on).	≈ 10 /arm
Engagement (satellites, snap-frozen)	pERK/total ERK in periventricular tissue (FGF17 gate anchor) and hippocampus; rapid ice-cold dissection (<2 min), phosphatase inhibitors (NaF, orthovanadate, β -glycerophosphate), positive-control lysate (FGF2-stimulated cells) and pre-specified assay CV on every blot; FGF17 at 15, 30 and 60 min; 7,8-DHF at 45 min after the 7th daily dose (descriptive). Reads pathway engagement, not TrkB specificity.	6 per group

7. Statistics and pre-registration

- Register on OSF before surgery (sponsor does this); ARRIVE 2.0 [14]; flow diagram of enrolled, excluded and completed animals.
- Estimation only: unadjusted contrasts with 95% intervals from a bootstrap that resamples within arm (BCa), reported beside the t-based interval; no Holm adjustment because no significance claims are made. Interaction exploratory.
- Sensitivity analyses: complete cases; index without the water maze (its own operating characteristics: false-go $\approx 10\%$ under no effect); set-shifting alone; partner identity described only (6 partners per environment cannot support a random effect).
- Sizing of the next study: use the upper 80% confidence limit of the pilot SD and the full factorial/interaction design; the "64 per arm for 0.5 SD" figure applies to a two-group comparison only.

8. Pre-registered decision rules and operating characteristics

Phase	Pass condition			
0a – reagents	FGF17 bioassay active on the first vial; mouse-rat alignment documented; DHF solution stable.			
0a – FGF17 engagement (gate)	Periventricular pERK/total ERK ≥ 1.5 -fold vehicle at ≥ 1 of 15/30/60 min (n=6 per group; assay CV pre-specified; positive-control lysate on every blot). If not met, the main cohort does not start; test a higher dose or continuous infusion in a small follow-up.			
0a – DHF engagement	Reported descriptively, not a gate (n=6, assay CV likely exceeds the effect). If not met, the DHF component is reported as "effect of unestablished mechanism".			
0b – feasibility	$\geq 50\%$ of the 64 enrolled rats (≥ 32) complete all three tasks; cannula placement $\geq 75\%$ correct (planned completion 62–69%; $P(\geq 32/64 \text{ complete}) = 0.82$ at 55%, 0.96 at 60%, 0.995 at 65%).			
0b – direction	$D+E - C \geq +0.4$ per-task SD ($\approx +0.55$ index-SD only at inter-task $r = 0.3$; 0.58 at $r = 0$; 0.86 at $r = 0.6$) <u>and</u> $D+E \geq$ both E and D in point estimate; interval reported, not gated.			
Provisional go	Phase 0a passed AND 0b feasibility AND 0b direction.			
0c (deferred add-on) – confirmed go	EdU+/Olig2+ density in corpus callosum higher in D+E than C (direction). If 0c is not run, the decision stays provisional.			
True situation (n=10/arm)	r=0: direction	r=0.3: direction	r=0.6: direction	r=0.3: + EdU direction*
No effect	5%	8%	10%	4%
Modest (E 0.3, D 0.1, D+E 0.4)	39%	37%	36%	27%
Target (E 0.3, D 0.2, D+E 0.6)	70%	61%	56%	53%
Additive (E 0.3, D 0.3, D+E 0.6)	68%	59%	54%	51%
Large (E 0.5, D 0.3, D+E 1.0)	97%	90%	85%	78%
Enrichment alone (E 0.5, D+E 0.5)	40%	38%	37%	19%

*EdU direction simulated as independent with a true drug effect of 0.5 SD where a drug effect exists. The rule is lenient and **cannot separate synergy from additivity** (61% vs 59%); it only screens for a promising direction. A "go" justifies the larger study only; a "no-go" identifies the failure mode (delivery, engagement, ceiling, attrition) and is not evidence that the approach cannot work.

9. India logistics and budget

Item	What the review found
Ethics	Rats are small animals: IAEC approval is sufficient (no central CPCSEA forwarding). CPCSEA requires IAEC meetings at least every 6 months (quarterly for larger programmes); proposals are due ≈ 30 days ahead; ask for an interim review.
Imports	FGF17 and antibodies are imported: 6–12 weeks, customs-house-agent fees ($\approx$ INR 5–15k per consignment) and dry-ice courier for each of at least two shipments (vial-first FGF17 order; later orders). Notification 51/96 gives $\approx 5\%$ duty on "scientific instruments, apparatus, equipment and spare parts" for DSIR-registered importers; coverage of reagents and any GST exemption are unconfirmed and the importing institute itself must hold DSIR registration. Without it, landed cost is ≈ 1.3 – $1.5\times$ list. Duty/GST rates are reviewers' knowledge, not verified by search.
Equipment access (precondition)	The budget assumes written, free access to: stereotaxic frame, water maze + tracking software, set-shift and maze apparatus, cryostat, fluorescence microscope and stereology software, western blot system. Hourly facility charges are NOT in any scenario; without written access the budget is void.
Timeline	IAEC 1–6 months; procurement 2–3 in parallel; Phase 0a 1–2; in-life ≈ 3 –4 (wave plan section 5); analysis 1–2; Phase 0c later. About 8–14 months from submission to core results.

Budget (USD; list prices from search plus reviewer knowledge; to be replaced by quotes):

Line (core = Phase 0a + 0b)		Base	Notes
Rats: 124 incl. satellites, partners, pilots, with transport		0.9k	INR 300–500 each
Housing/care ≈ 67 –70 days per main-cohort rat; partners longer		1.5k	per-diem not found; optimistic 0.75k
FGF17 3 $\times$ 25 μ g, landed		1.0k	Cohesion $\approx$ USD 480/50 μ g via an aggregator; R&D; 25 μ g mouse vial price not found; PeproTech lists human only; 4th vial 0.45k in pessimistic
7,8-DHF ≈ 4 g, landed		0.35k	TCI $\approx$ USD 62/g
Cannulas, dummies, screws, cement (94 rats)		0.5k	
Anaesthesia, analgesia, surgical/terminal consumables (94 cannulations + 100 terminal procedures)		1.0k	
ELE arena, wheels, objects, puzzle devices (18% GST on fabrication)		0.65k	
Behaviour consumables and fabrication (set-shift pots/media/odours, Hebb-Williams maze) incl. GST		0.8k	
EdU (banks the label)		0.2k	
ERK block, landed: pERK, total ERK, HRP secondaries, loading control, FGF2 positive control		1.55k	list ≈ 1.15 k
Western reagents (membranes, ECL, inhibitors)		0.45k	
Two shipments with customs-agent fees		0.6k	
IAEC fee and miscellaneous		0.1k	
Core base total (no reserve)		9.6k	
Phase 0c add-on (deferred): EdU click kit + Olig2 + secondary landed 1.34k, IHC reagents 0.25k, shipping 0.1k		1.7k	CC1 ($\approx$ USD 594 list) and PSD-95/synaptophysin westerns are further add-ons
Full programme base		11.3k	
Scenario	Core	Full	Assumptions
Optimistic	≈ 8.1 k	≈ 9.5 k	DSIR concession (antibodies at list, one cheaper shipment, FGF17 0.85k); low per-diem (housing 0.75k)
Base	≈ 9.6 k	≈ 11.3 k	as above
Pessimistic	≈ 11.5 k	≈ 13.4 k	15% overrun plus a 4th FGF17 vial; no DSIR; equipment hourly fees not included

Phases: 0a ≈ 3.7 k (satellite rats, first FGF17 vial, ERK antibodies, surgery, first shipment) with a stop rule at that spend; 0b ≈ 5.9 k; 0c ≈ 1.7 k deferred. **First cuts if prices run high:** enrol 14 per arm; drop the 60-min satellite group; run 0c later. The western block must not be dropped (it carries the only gate). **Must not be cut:** dye/histology placement check, FGF17 bioassay and alignment, blinding/randomisation, DHF vehicle tolerability check, visible-platform control, banking of tissue.

10. Lab handover and remote oversight

The researcher will not be on site; the host laboratory runs the experiment. The PDF is therefore the working protocol, and the following must exist before animals arrive.

Item	Responsibility / requirement
Roles	Host principal investigator (accountable for IAEC compliance and animal welfare); study coordinator (day-to-day, one named person); ≥ 3 trained operators (ELE/dosing; two testing) plus a surgeon; sponsor (design, randomisation, drug-code key, registration, independent analysis).
Agreements	Written MoU/collaboration agreement covering funding, overhead (institutes often charge overhead on external funds), payment schedule, data and tissue ownership, authorship, publication, and stop rules. Foreign-funds rules for receipt of money by Indian institutions (e.g. FCRA) must be checked with the institute's accounts office; not verified here.
SOP pack (host completes with photos; sponsor reviews)	Surgery and cannula care; ICV bolus injection; vehicle preparation and dosing; ELE and control sessions (arena, objects, puzzle ladder, rewards, logs); set-shifting; Hebb-Williams; water maze incl. temperature/hypothermia rules; food restriction and weighing; euthanasia, hemisection and tissue banking; satellite dissection and western protocol; per-day time-budget table; welfare scoresheet.
Blinding and randomisation	Sponsor generates the allocation, labels coded vials and arm IDs, and keeps the key; host staff handle coded IDs only; unblinding after data lock.
Data and oversight	Video of all sessions under coded IDs; raw data sheets in a fixed template; weekly summary (enrolment, weights, exclusions, adverse events, deviations log) and a photo/video check-in; raw data and images shared with the sponsor for independent blinded analysis. Gates (section 8) are decided by the sponsor from the shared data; any deviation is recorded.
Reagent chain	Sponsor/host agree quotes; vial-first FGF17 order; bioassay result and sequence alignment shared before the main cohort starts; cold-chain receipt log.
Go/no-go communication	After 0a and 0b the sponsor issues a written decision (provisional go / stop / redesign) before further spending.

11. Validation record for this adaptation

Round	Reviewers	Verdicts	Main issues → fix
Earlier A	3 Opus (design; pharmacology; logistics)	All approve with changes	Pump ends before testing; pump cost; middle-aged rats; tissue conflict → boluses, young adults, new sourcing.
Earlier B	3 Opus + 1 Haiku; user changed the aim	All approve with changes	Transfer overlap; social confound; composite not a g factor; attrition; timeline; budget.
Translational	1 Opus	Approve with changes	Overclaiming; neurotrophin history; FGF safety; paediatric ethics.
Round 1 of 3	5 Opus + 2 Sonnet	All 7 approve with changes	Index scale and gate miscalibration; BrdU/Olig2 on adjacent sections; phospho pre-analytics; timelines; ELE trained set-shift dimensions; budget overhead; 25 inconsistencies.
Round 2 of 3	5 Opus + 2 Sonnet	All 7 approve with changes	"Go" rule not a real gate and cannot separate synergy from additivity; wheel/weight covariates aliased; complete-case bias; no frozen tissue from the learning cohort; engagement satellites underpowered (n=4); transfer mislabelled; Hebb-Williams reliability; wave workload; shipping vs vial-first; equipment fees; refeed overlap; 17 inconsistencies → all applied in this version.
Round 3 of 3	5 Opus + 2 Sonnet	(running)	This version (v3) is under review.

12. Items the reviewers still do not like / remaining gaps (after round 2 of 3)

#	Gap	Kind
1	Bolus FGF17 is not the Iram infusion ($\approx 7\times$ less volume); dose unanchored until the Methods are read; null confounded with delivery.	Pumps in the larger study
2	Power: only very large effects detectable; healthy rats near ceiling; the direction rule is lenient and cannot separate synergy from additivity.	Intrinsic to a pilot
3	7,8-DHF may be inert in vivo; 53 daily injections extrapolate beyond 4 weeks of rat precedent; pERK shows pathway engagement only; DHF gate is descriptive.	Declared
4	ELE confounds social contact, exercise, puzzles and spatial experience; environment arm cannot be fully blinded; set-shifting is the only clean transfer readout.	Declared
5	Cumulative welfare burden may be challenged by the IAEC; attrition may exceed the plan.	Pre-review requested
6	Prices unverified; equipment hourly fees excluded; DSIR coverage of reagents unconfirmed; pessimistic core 11.5k exceeds the cap.	Quotes needed
7	Staffing: three operators needed for overlapping waves; otherwise ≈ 17.5 weeks.	Host to confirm
8	Structure (EdU/Olig2) is deferred, so the provisional go lacks a structural marker; no synaptic or neuroplasticity readout in this study.	Phase 0c / Stage 1
9	Optogenetics Phase 2 and the human roadmap are aspirations without design detail or validated tools in the host lab.	Own review round
10	Literature checks used search snippets only; several references partial; human-trial and neurotrophin statements to be verified.	Researcher to verify

13. References and verification status

Checks were by search snippets only (PubMed, PMC, Crossref and publisher pages unreachable). "Verified" means consistent across indexed records.

#	Citation	Status
1	Iram T, Kern F, Kaur A, et al. Young CSF restores oligodendrogenesis and memory in aged mice via Fgf17. <i>Nature</i> 2022;605:509–515. doi:10.1038/s41586-022-04722-0	Verified; delivered dose not retrieved
2	Boltaev U, et al. Multiplex quantitative assays indicate a need for reevaluating reported small-molecule TrkB agonists. <i>Sci Signal</i> 2017;10(493). doi:10.1126/scisignal.aal1670	Verified
3	Todd D, Gowers I, Dowler SJ, et al. A monoclonal antibody TrkB receptor agonist as a potential therapeutic for Huntington's disease. <i>PLoS One</i> 2014;9(2):e87923. doi:10.1371/journal.pone.0087923	Citation verified; DHF content via reviews
4	Do small molecules activate the TrkB receptor in the same manner as BDNF? <i>Pharmaceuticals</i> 2021;14(8):704 (authors not captured).	Verified (review)
5	Rawlings-Mortimer F, et al. 7,8-DHF during water-maze training in mice. <i>Front Syst Neurosci</i> 2023. doi:10.3389/fnsys.2023.1134594	Partial
6	van Praag H, Kempermann G, Gage FH. <i>Nat Rev Neurosci</i> 2000;1:191–198. doi:10.1038/35044558	Verified
7	Nithianantharajah J, Hannan AJ. <i>Nat Rev Neurosci</i> 2006;7:697–709. doi:10.1038/nrn1970	Verified
8	Galsworthy MJ, et al. <i>Behav Genet</i> 2005;35:675–692. doi:10.1007/s10519-005-3423-9	Verified
9	Matzel LD, Han YR, Grossman H, et al. <i>J Neurosci</i> 2003 (general learning ability in mice)	Partial
10	Gibson EM, Purger D, Mount CW, et al. Neuronal activity promotes oligodendrogenesis and adaptive myelination in the mammalian brain. <i>Science</i> 2014;344:1252304 (optogenetic precedent)	To be verified
11	Birrell JM, Brown VJ. <i>J Neurosci</i> 2000;20(11):4320–4324 (set-shifting); Rabinovitch MS, Rosvold HE. <i>Can J Psychol</i> 1951;5(3):122–128 (Hebb-Williams)	Birrell verified; Rabinovitch partial
12	Kraemer HC, et al. <i>Arch Gen Psychiatry</i> 2006;63:484–489. doi:10.1001/archpsyc.63.5.484	Verified
13	Leon AC, Davis LL, Kraemer HC. <i>J Psychiatr Res</i> 2011;45:626–629. doi:10.1016/j.jpsychires.2010.10.008	Verified
14	Percie du Sert N, et al. <i>PLoS Biol</i> 2020;18:e3000410. doi:10.1371/journal.pbio.3000410	Verified
15	Enrichment meta-analysis (rodent cognition $g \approx 0.44$, 95% CI 0.25–0.62): Univ. at Albany dissertation (scholarsarchive.library.albany.edu/etd/371)	Partial; not peer-reviewed
16	Léger M, et al. 2014 (duration of enrichment effects in mice)	Partial

Not found: any study of FGF17/FGF-family treatment improving learning in healthy young rodents; any study combining 7,8-DHF with enrichment; rat ICV FGF17 doses; Iram 2022 delivered dose; Indian prices for Alzet pumps and recombinant FGF17; a peer-reviewed validation of a Hebb-Williams subset; per-diem housing cost.